Engineering Applications of Artificial Intelligence 158 (2025) 111482

Contents lists available at ScienceDirect

Engineering Applications of Artificial Intelligence

journal homepage: www.elsevier.com/locate/engappai

Long short-term memory neural network- *decline curve analysis* production forecast method for horizontal wells in tight reservoir based on sequence decomposition and reconstruction

Jinxin Cao [a,b], Yiqiang Li [a,b,*], Yaqian Zhang [a,b], Xuechen Tang [a,b], Qihang Li [a,b], Yuling Zhang [a,b], Zheyu Liu [a,b,**]

[a] *National Key Laboratory of Petroleum Resources and Engineering, China University of Petroleum, Beijing, 102249, China*
[b] *College of Petroleum Engineering, China University of Petroleum, Beijing, 102249, China*

ARTICLE INFO

Keywords:
Tight oil
Production forecast
Empirical mode decomposition
Long short-term memory
Decline curve analysis
Physical constraints

ABSTRACT

Oil production is a key parameter for evaluating geological potential. Conventional methods struggle to forecast nonlinear and non-stationary production sequences with a strong temporal trend due to the poor physical properties of tight reservoirs. This research proposes a forecast method that integrates a long short-term memory (LSTM) neural network with decline curve analysis (DCA). First, empirical mode decomposition (EMD) is applied to the production sequence, extracting multiple fluctuating intrinsic mode functions (IMF) related to manual construction and a residual (RES) representing reservoir energy depletion. Approximate entropy (ApEn) is then used to categorize each IMF into three groups based on sequence decomposition results, facilitating the reconstruction of the production sequence. LSTM forecasts the recombined IMF sequence, while DCA predicts the residual component. Results indicate that EMD effectively separates time trends, and both reconstructed components and residuals can be accurately predicted. Compared with stand-alone LSTM, back-propagation (BP) neural network, random-forest (RF) and convolutional-neural-network (CNN) models, the proposed method reduces production forecast errors by at least 25 %. This research incorporates signal-processing techniques and physical constraints into production forecasting. These enhancements provide a more accurate and reliable method for forecasting production in hydraulically fractured horizontal wells within tight-oil reservoirs.

1. Introduction

With the continued exploration and development of oil fields, global efforts have shifted toward maintaining stable production of conventional oil while accelerating the growth of unconventional oil. Tight oil reservoirs, a key type of unconventional oil, with abundant reserves and have become a hot spot in exploration research (You et al., 2025; Abdrabou et al., 2025). According to EIA data, as of 2021, tight oil accounted for more than 60 % of total U.S. oil production. Projections indicate that over the next 30 years, tight oil will remain the dominant source of oil production in the United States (Fig. 1).

Oil production is the key parameter for evaluating geological potential, formulating oil well production plan and adjusting oil well production system. Accurate production prediction is of great significance for oil fields (Davtyan et al., 2020). Common production forecasting methods include empirical formulas (Sharma et al., 2019), analytical models (Wu et al., 2019; Shi et al., 2015; Wang et al., 2023), and numerical simulations (Myshakin et al., 2016). Compared to conventional reservoirs, tight reservoirs exhibit lower porosity and permeability, resulting in rapid production decline and greater sensitivity to external factors (Tugan and Weijermars, 2020). Additionally, large-scale fracturing in tight reservoir wells, combined with complex subsurface conditions, complicates production patterns. Conventional forecasting methods are no longer sufficient for accurately forecasting the production of horizontal wells (Li et al., 2022).

In recent years, the widespread application of artificial intelligence in the fields of science and engineering has driven digital transformation in the oil development industry (Nadji-Tehrani and Eslami, 2020; Wang et al., 2009; Ling et al., 2019; Feng et al., 2023; Das et al., 2025). This trend has significantly advanced research on oilfield production

* Corresponding author. National Key Laboratory of Petroleum Resources and Engineering, China University of Petroleum, Beijing, 102249, China
** Corresponding author. National Key Laboratory of Petroleum Resources and Engineering, China University of Petroleum, Beijing, 102249, China
E-mail address: yiqiangli@cup.edu.cn (Y. Li).

https://doi.org/10.1016/j.engappai.2025.111482
Received 18 April 2024; Received in revised form 1 June 2025; Accepted 9 June 2025
Available online 11 June 2025

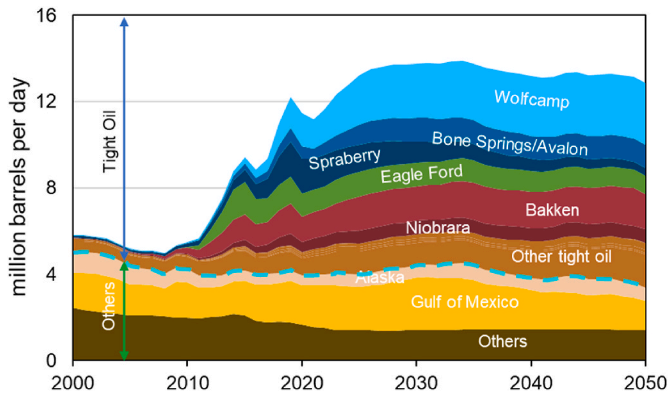


Fig. 1. U.S. oil and gas annual production forecast (EIA, 2021).

forecasting methods (Ning et al., 2022; Chahar et al., 2022; Wang et al., 2023; Yan et al., 2025). Machine learning, with its strong nonlinear fitting capability (Chalmond and Girard, 1999; Cheng and Bai, 2021), can bypass many complex nonlinear mapping relationships in oilfield development and production. Research in this area can be categorized into two approaches: single models and hybrid models. Due to the "black box" nature of machine learning models, concerns about interpretability persist among researchers (Hakkoum et al., 2024). In addition, although extensive research has been conducted on production forecasting, most studies focus on conventional reservoirs with intermediate to high permeability (Ng et al. 2022). In contrast, tight reservoir production exhibits significant fluctuations, and long-term trends driven by natural energy depletion are more pronounced. This poses a challenge for purely data-driven machine learning methods, which lack physical constraints.

This research considers oil well production data include two components: natural depletion and fluctuations caused by external factors. The natural depletion component represents a long-term trend that is challenging for machine learning to forecast accurately. Additionally, due to various influencing factors, the time-frequency relationship of tight oil well production is highly complex and exhibits strong fluctuation. To address these challenges, this paper proposes an LSTM-DCA forecasting method based on signal decomposition and reconstruction. As shown in Fig. 2, the production data is first decomposed using EMD method to extract several *IMFs* and a RES. Then, the data is reconstructed into several relatively simple sequences based on the *ApEn* of the *IMFs*. The reconstructed sequences are forecasted using LSTM, while the separated RES component is analyzed by DCA.

Our findings confirm the effectiveness of signal processing methods in forecasting the production sequences of oil wells. The EMD method effectively decomposes the production sequence into a RES component and several *IMFs*. K-means clustering based on approximate entropy reduces computational costs while maintaining forecast accuracy. By explicitly separating physics-governed depletion from data-driven fluctuations, our workflow retains the interpretability engineers rely on for operational decisions. The results of this research offer new insights for time series forecast research.

The remainder of this paper is organized as follows: Section 2 reviews related work. Section 3 introduces the characteristics and preprocessing methods of the data used in this research, followed by a detailed explanation of the methods incorporated into the proposed model framework. Section 4 presents the validation and analysis of the proposed model. Section 5 emphasizes the key findings and contributions of this research, while Section 6 provides a conclusion and discusses future works.

2. Related works

The forecast of oil well production includes traditional physics-driven methods, data-driven machine learning methods, and hybrid methods.

2.1. Physics-driven methods for production forecast

The commonly used production forecasting methods include empirical formulas, analytical models, and numerical simulations. Empirical methods, such as the Arps decline curve, rely on historical production data and assumed decline patterns to forecast future output. However, they cannot capture fluctuations, and due to varying reservoir conditions, decline patterns differ significantly, limiting their accuracy (Jongkittinarukorn et al., 2023). Reservoir conditions are highly complex, and oil well production depends on large datasets that often exhibit strong nonlinearity and contain missing values. Under these circumstances, developing precise analytical models is nearly impossible (Cao et al., 2022). Similarly, numerical simulation methods are affected by uncertainties in subsurface conditions due to incomplete data.

The effectiveness of these methods is constrained by unknown underground conditions, challenges in accurately characterizing flow patterns, and high time and computational costs.

2.2. Data-driven machine learning methods for production forecast

The rapid advancement of machine learning has led to faster application of data-driven methods for forecasting oil well production. Early forecasting primarily relied on traditional machine learning techniques (Bhattacharya et al., 2019). analyzed key factors influencing daily gas production using a spatio-temporal oilfield database and compared the forecast performance of support vector machines (SVM), random forests (RF), and artificial neural networks (ANN). Their findings indicated that ANN offer high forecast accuracy and computational efficiency (Zhang et al., 2022). applied a BP neural network to gas production forecasting and found that a single-layer neural network outperformed a two-layer network in fitting accuracy. With the proposal of complex models such as recurrent neural networks (RNN), deep neural networks (DNN), and

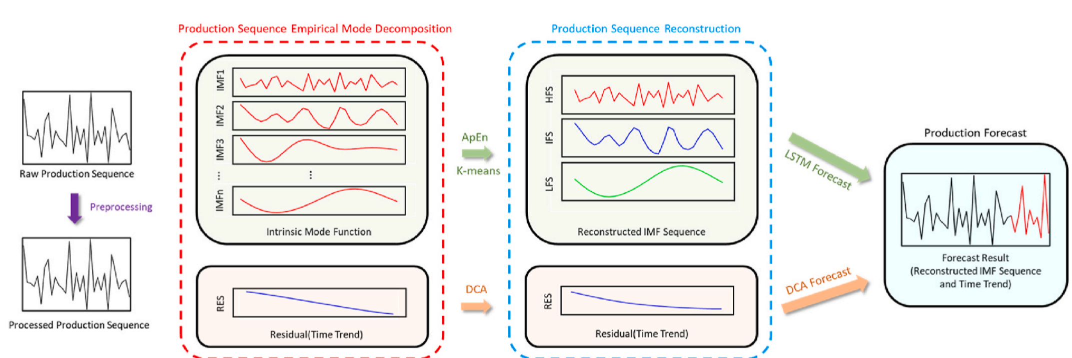


Fig. 2. Flow chart of tight oil horizontal wells production forecast based on sequence decomposition and reconstruction.

convolutional neural networks (CNN), numerous AI-driven production forecasting methods have emerged. Well-trained DNN models have been successfully integrated into hydraulic fracturing designs to estimate 6-month and 18-month cumulative oil and gas production demonstrated that the forecast performance of DNN models is significantly influenced by factors such as the activation function, number of hidden layers, and number of neurons (Wang and Chen, 2019). Researchers have also explored ensemble models (Kumar et al., 2023). incorporated an attention mechanism into a LSTM neural network to improve production forecast accuracy. With the introduction of the Transformer model, time series forecast research has reached its peaks (Yan et al., 2025). proposed a Transformer-based approach for forecasting pre-extraction parameters and completion time in coalbed methane production.

While these advancements have enhanced model performance, they often emphasize complex architectures with large parameters and focus on improving individual models. This increases training costs and reduces interpretability.

2.3. Hybrid methods for production forecast

Although machine learning has significantly improved the forecast accuracy of production, its results remain unsatisfactory in some cases. To address this, researchers have begun integrating multiple machine learning models (Zhou et al., 2023). proposed a model combining a CNN and a bidirectional gated recurrent unit (BiGRU) with an attention mechanism to enhance shale oil production forecasting accuracy (Wen et al., 2023). developed a hybrid model integrating LSTM and support vector regression (SVR) for oil production forecast (Zha et al., 2022). designed a model that employs CNN for feature extraction and LSTM for time series forecasting. Recognizing that oil well production exhibits unpredictable temporal trends, researchers have introduced preprocessing techniques before making predictions (Fan et al., 2024). applied a DNN to forecast ore production using truck haulage data and weather conditions in open-pit mines. Prior to building the model, principal component analysis (PCA) was used to reduce data dimensionality and mitigate multicollinearity among highly correlated input variables (Fan et al., 2021). developed an approach based on production time and LSTM-ARIMA, achieving more accurate oil production forecasts (Fragalla et al., 2025). proposed a method that preprocesses data using multilevel discrete wavelet decomposition, followed by a Transformer model with a multi-head attention mechanism for natural gas production modeling.

Recent research has increasingly focused on integrating multiple models to capitalize on their respective strengths. However, research on interpretable models constrained by physical principles remain limited. Notably, the mathematical characterization of flow processes in tight oil reservoirs remains debated and unresolved. Therefore, this research incorporates an empirical formula-based model as a physical constraint.

3. Method

3.1. Data features and preprocessing

This research analyzes production data from a fractured horizontal well in a tight oil block. Due to certain factors, the well was temporarily shut in, leading to data loss. To address this, cubic spline interpolation was applied to reconstruct the missing data. Finally, the 886-day production sequence of the well was obtained, and its characteristics are shown in Table 1.

Table 1
Statistical characteristic of production data.

Parameter	Volume	Mean Value	Variance	Standard Deviation	Approximate Entropy
Value	1400	8.2365	35.4640	5.9552	1.4903

The preprocessed production sequence is shown in Fig. 3. In the initial production stage, the natural energy of the reservoir is depleted, and the production continues to decline rapidly. At this time, production fluctuates significantly, as highlighted in red in Fig. 3. As natural energy continues to deplete, production stabilizes at a low level, as shown in the blue in Fig. 3.

3.2. Empirical mode decomposition (EMD)

To address the challenge of defining instantaneous frequency (Huang et al., 1998), introduced Empirical Mode Decomposition (EMD). EMD is a time-frequency analysis technique designed for nonlinear and non-stationary signals. Based on the characteristics of the input data, the method decomposes the signal into several narrow-band components, the intrinsic mode functions (*IMFs*) and a residual component.

$$Q(t) = \sum_{i=1}^{n} IMF_i(t) + RES(t) \tag{1}$$

where Q is the production of a certain oil well; *IMFi* is the *i*th intrinsic mode function obtained by decomposition and *RES* is the residual component.

Each *IMF* component must satisfy two fundamental conditions: on the entire sequence, the difference between the number of extreme points and the number of zero-crossing points is not more than 1; at any point, the mean value of the upper and lower envelopes is 0.

According to the *IMF* definition and the production characteristics of tight oil wells. *RES* captures the intrinsic production decline trend over time, while the *IMF* components reflect production fluctuations caused by external factors such as manual operations, including well switching. Each *IMF* can be extracted by the following method:

First, find all $Q_+(t)$ in $Q(t)$, and fit $Q_+(t)$ through the cubic spline function; similarly find $Q\text{-}(t)$. Calculate $Q_{m1}(t)$ of the original production sequence (Wang et al., 2009).

$$Q_{m1}(t) = \frac{Q_+(t) + Q_-(t)}{2} \tag{2}$$

where $Q_+(t)$ is the maximum value points of the original production sequence; $Q\text{-}(t)$ is the minimum value envelope; $Q_{m1}(t)$ is the mean value of the maximum and minimum value envelope.

Subtract $Q_{m1}(t)$ from the original sequence to get $Q_{h1}^1(t)$.

$$Q_{h1}^1(t) = Q(t) - Q_{m1}(t) \tag{3}$$

where $Q_{h1}^1(t)$ is a new production sequence.

In most cases, $Q_{h1}^1(t)$ does not immediately satisfy the *IMF* conditions. The above steps must be repeated until the definition is met. Assuming that it is satisfied after the k iterations, the first *IMF* component is:

$$IMF_1(t) = Q_{h1}^k(t) \tag{4}$$

where $Q_{h1}^k(t)$ is the sequence obtained by the kth calculation.

However, it is difficult to satisfy the mean value of the upper and lower envelopes of the *IMF* to be 0. It is generally considered that when *SD* coefficient is less than ε, 0.25 is selected in this study.

$$SD = \frac{\sum \left[Q_{h1}^{k-1}(t) - Q_{h1}^k(t)\right]^2}{\sum \left[Q_{h1}^{k-1}(t)\right]^2} \le \varepsilon \tag{5}$$

Where ε is the screening threshold; *SD* is the standard deviation.

This sequence is a high-frequency components, and a new sequence $Q_{r1}(t)$ can be obtained by subtracting $IMF1(t)$ from $Q(t)$.

$$Q_{r1}(t) = Q(t) - Q_{h1}^k(t) \tag{6}$$

Where $Q_{r1}(t)$ is the sequence obtained by original sequence minus *IMF1*.

This process is iteratively applied to $Q_{r1}(t)$ to extract $IMF2(t)$ and

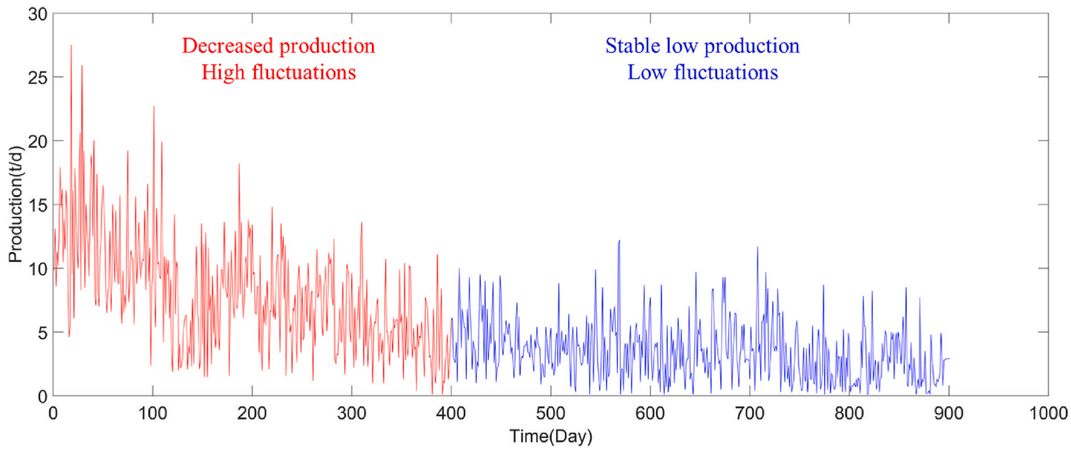


Fig. 3. Production data of horizontal wells in a tight oil block.

$Q_{r2}(t)$, and so on until $Q_{rn}(t)$ under at nth order *IMF* component satisfies certain rules and stops.

3.3. Sequence reconstruction based on approximate entropy

During EMD, many *IMFs* can often be obtained. If each *IMF* component is forecasted separately, cumulative errors may arise, leading to significant deviations in the final forecast and the time costs will increase. Therefore, it is necessary to combine *IMFs* with similar characteristic parameters to reconstruct production sequence.

Accurate entropy calculations require a large amount of data by modifying the exact regularity statistic *Kolmogorov-Sinai entropy*. The results will be limited by the extreme influence of system noise (Hornero et al., 2005). Approximate entropy (*ApEn*) is a nonlinear analysis technique used to quantify the regularity and unpredictability of time series data fluctuations (Pincus and S., 1991).

To compute *ApEn*, first, a window of length m is defined to generate a set of m-dimensional vectors:

$$\overline{IMF}(t) = [IMF(t), IMF(t+1), \ldots, IMF(t+m-1)] \tag{7}$$

where $\overline{IMF}$ is the intrinsic mode function.

Among them, $t = 1 \sim T\text{-}m+1$, then there is

$$\overline{IMF}(t) = [IMF(t), IMF(t+1)] \tag{8}$$

Use the value of each $\overline{IMF}$ in the above formula to calculate the distance between the vector $\overline{IMF}$ and the remaining vectors $\overline{IMF}$ *(k)* on the time sequence

$$d[\overline{IMF}(t), \overline{IMF}(k)] = \max_{l=0\sim m-1} |IMF(t+l) - IMF(k+l)| \tag{9}$$

Which represents the maximum difference between the corresponding elements of vector $\overline{IMF}$ *(t)* and $\overline{IMF}$ *(k)*.

Given a predefined similarity tolerance r threshold, count d $\overline{IMF}$ *(t)*, $\overline{IMF}$ *(k)*] $< r$ number and total vector number $T\text{-}m+1$ corresponding to each t value, denoted as $C_t^m(r)$ has

$$C_t^m(r) = \frac{\{d[\overline{IMF}(t), \overline{IMF}(k)] < r, \text{number}\}}{T-m+1} \tag{10}$$

Applying a logarithmic transformation to on $C_t^m(r)$, and then calculate the average value of all t, denoted as

$$\phi^m(r) = \frac{1}{T-m+1} \sum_{t=1}^{T-m+1} \ln C_t^m(r) \tag{11}$$

Increase m to $m+1$ and repeating the above steps produces $\Phi^{m+1}(r)$, then there is

$$ApEn_{IMF}(m, r, T) = \phi^m(r) - \phi^{m+1}(r) \tag{12}$$

To enhance forecasting accuracy, reduce errors, and improve computational efficiency, *IMFs* with similar approximate entropy values are merged while preserving the essential characteristics of the production sequence. In this study, the K-means clustering algorithm is applied to categorize the *IMFs* into three groups based on approximate entropy. Three combined components: high frequency sequence (*HFS)*, intermediate frequency sequence (*IFS)* and low frequency sequence (*LFS)* were formed.

$$HFS = \sum_{i=1}^{a} IMF_i, IFS = \sum_{i=a+1}^{a+b} IMF_i, LFS = \sum_{i=a+b+1}^{a+b+c} IMF_i \tag{13}$$

where *a, b* and *c* represent the number of *IMFs* assigned to *HFS*, *IFS*, and *LFS*, respectively.

3.4. Long short-term memory neural network

To address the issues of gradient vanishing and explosion in standard RNN, LSTM was proposed. Unlike RNN, which have only a hidden state h, LSTM introduces a cell state c to store long-term information.

The structure of LSTM is illustrated in Fig. 4. Three gate structures (forget gate, input gate, and output gate) are employed to control the long-term state c. The forget gate determines the retention degree of $ct-1$ from the previous moment to the current moment ct, the input gate determines the retention degree of the input xt of the current network, and the output gate determines the output from ct to ht (Domala and

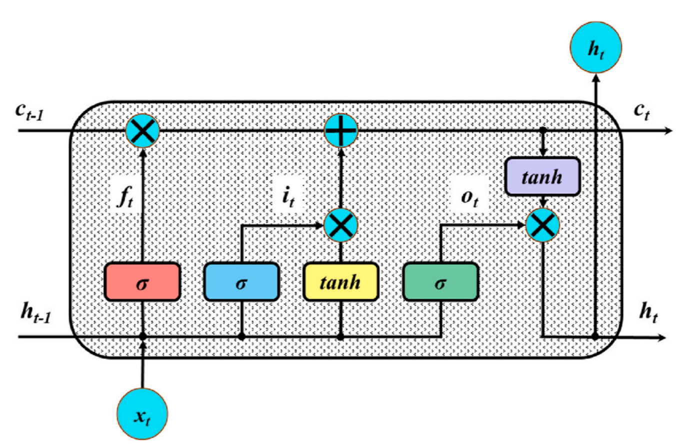


Fig. 4. Schematic diagram of LSTM structure.

Kim, 2023).

In the forget gate calculation,

$$f_t = \sigma\left(W_f \cdot [h_{t-1}, c_t] + b_f\right) \tag{14}$$

where W_f is the weight matrix of the forget gate; $[h_{t-1}, c_t]$ represents the concatenation of the hidden state and input; b_f is the bias item of the forget gate; σ is the sigmoid activation function.

In the input gate calculation,

$$\begin{aligned} i_t &= \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \\ \widetilde{c}_t &= tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \end{aligned} \tag{15}$$

where W_i is the weight matrix of the input gate; b_i is the bias term of the input gate; $\sim ct$ is the candidate cell state.

The update rules of ct at the current moment are as follows,

$$c_t = f_t \times c_{t-1} + i_t \times \widetilde{c}_t \tag{16}$$

In the output gate calculation,

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{17}$$

LSTM final state output,

$$h_t = o_t * \tanh(c_t) \tag{18}$$

Using LSTM, accurate forecasts for tight oil production can be achieved by modeling three reconstructed sequences: *HFS*, *IFS* and *LFS*.

The LSTM network architecture is illustrated in Fig. 5. To prevent overfitting, a dropout layer is incorporated. The network consists of an input, an LSTM, a dropout, a ReLU activation, a fully connected, and a regression layer. This model uses the production data of the first 15 days to forecast the production data of the 16th day. Consequently, the input size is 15, the number of LSTM cells is 100, and the dropout probability is 0.2.

Before training, the data is normalized using min-max scaling, mapping it to the range [0,1].

$$q = \frac{Q - Q_{min}}{Q_{max} - Q_{max}} \tag{19}$$

where q is the normalized production, Q is the actual production, $Q_{\min}$ is the minimum production, $Q_{\max}$ is the maximum production.

During training, the Adam optimizer is used with a maximum of 1000 epochs. The initial learning rate is set to 0.005 and decays to 0.0005 after 800 epochs.

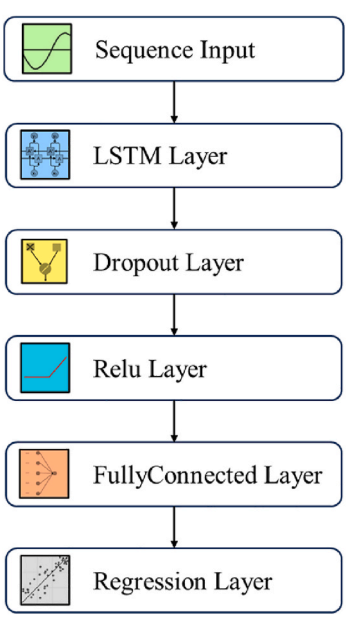


Fig. 5. LSTM neural network structure.

3.5. Decline curve analysis (DCA)

In this research, the Arps production decline curve is employed to analyze the variation pattern of RES. It is observed that RES follows Arps exponential decline model. The mathematical expression of Arps exponential decline curve is as follows [34]:

$$Q(t) = Q_i \times e^{-D_i t} \tag{20}$$

where Q_i and D_i are model fitting parameters, which is fitted by MATLAB Curve Fitting Tools.

3.6. Model evaluation method

Mean squared error (MSE) (Gonzalez-Zapata et al., 2023) is a widely used indicator to measure the difference between the forecasted value and the actual value. The MSE is obtained by calculating the average of the squares of the differences between the forecasted values and the actual values:

$$MSE = \frac{\sum_{i=1}^{N} (Q_i - \widehat{Q}_i)^2}{N} \tag{21}$$

where N is the data length; Q_i is the actual data; $\acute{}Qi$ is the forecasted result.

Root mean square error (RMSE) (Zheng and Hu, 2023) is similar to the MSE but takes the square root of MSE, providing a more interpretable error measurement in the same unit as the original data:

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N} (Q_i - \widehat{Q}_i)^2}{N}} \tag{22}$$

Mean absolute error (MAE) (Zheng and Hu, 2023) is another widely used indicator for evaluating model accuracy. It calculates the average absolute difference between forecasted and observed values, making it less sensitive to large errors than MSE:

$$MAE = \frac{1}{N} \sum_{i=1}^{N} |Q_i - \widehat{Q}_i| \tag{23}$$

Mean absolute percentage error (MAPE) (Najafzadeh and Mahmoudi-Rad, 2024) is similar to MAE but expresses the error as a percentage, making it independent of the data's scale:

$$MAPE = \frac{1}{N} \sum_{i=1}^{N} \left| \frac{\widehat{Q}_i - Q_i}{Q_i} \right| \tag{24}$$

4. Result and analysis

4.1. Production sequence decomposition

In this research, EMD method is applied for sequence decomposition, the number of extrema of the *RES* obtained by EMD is employed as the stop standard. The *RES* obtained with different numbers of extreme values is shown in Fig. 6.

As previously discussed, the production of fractured horizontal wells in tight oil reservoirs follows an initial rapid decline, followed by a slower decline phase. When the number of extrema is 1, the *RES* aligns with the production decline trend, indicating that the number of extrema has no significant effect on *RES*.

Five *IMFs* and one *RES* are extracted by EMD as shown in Fig. 7. From *IMF1* to *IMF5*, the sequence frequency gradually decreases. *RES* is basically in line with the production trend that begins to decline slowly at 400 days, and finally maintains a stable and low production state. The *ApEn* each *IMF* is calculated, and the results, shown in Fig. 8, reveal a

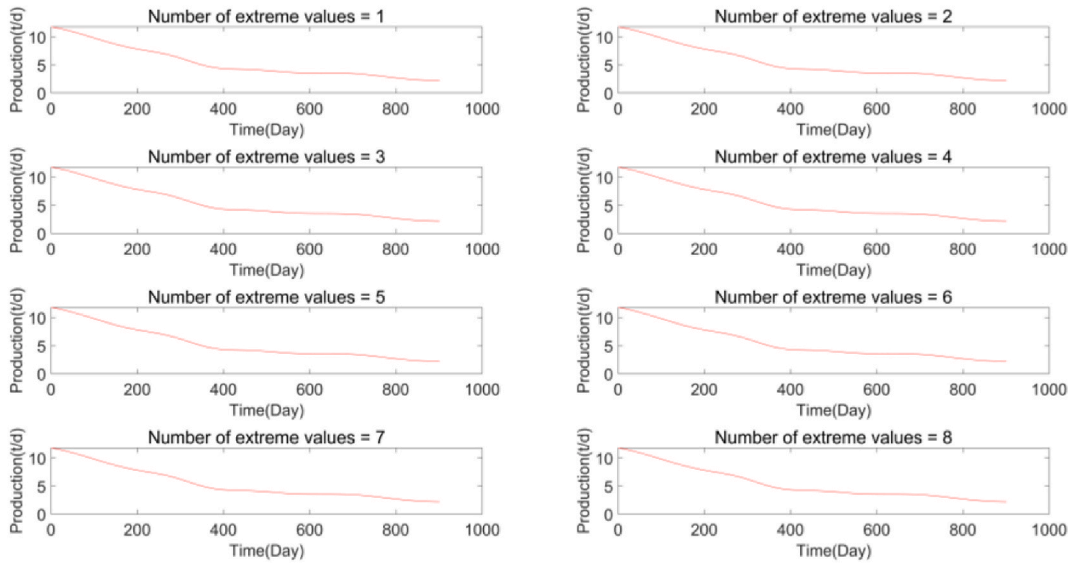


Fig. 6. Residual sequence obtained by EMD with different numbers of extrema.

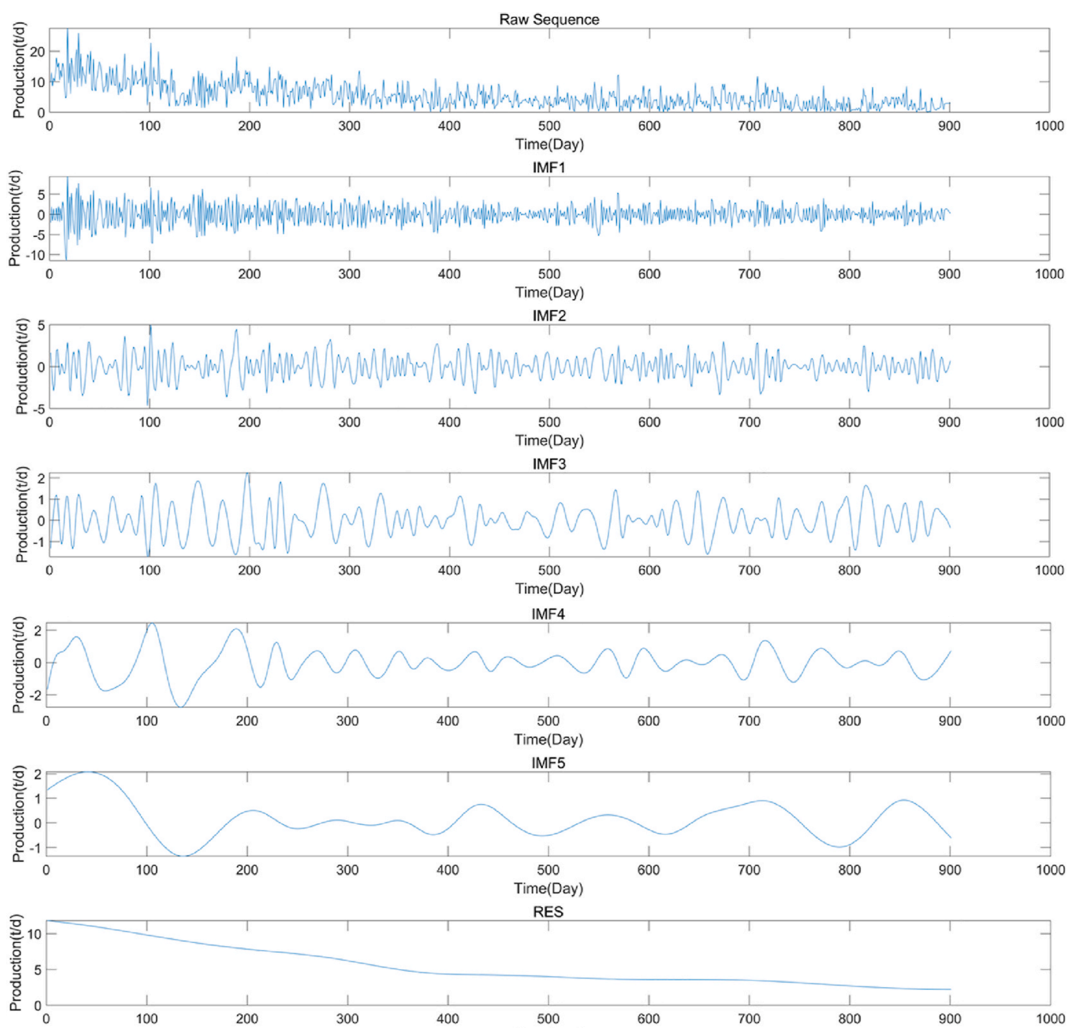


Fig. 7. EMD decomposition results of production sequence of horizontal production wells in tight reservoirs.

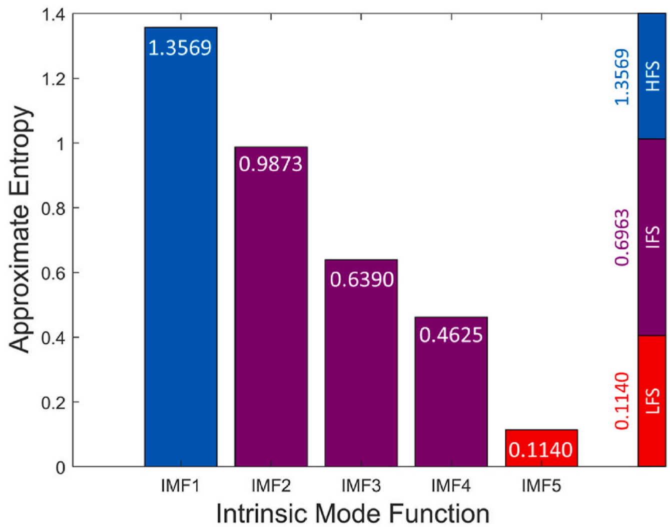


Fig. 8. *ApEn* calculation and K-means cluster results of *IMFs* in oil well production sequence.

decreasing trend from *IMF1* to *IMF5*, which indicates that high-frequency sequences contain more information than low-frequency sequences.

4.2. Production sequence reconstruction

To reconstruct the production sequence, K-means clustering is employed to the *ApEn* of each *IMF*, which is divided into HFS, IFS, LFS three categories. Therefore, the number of clusters is set to 3. The clustering result is shown in Fig. 8, where *IMF1* belongs to *HFS*, *IMF2-IMF4* belongs to *IFS*, and *IMF5* belongs to *LFS*. The cluster centers and boundaries are shown in Fig. 8. The relationship between the reconstructed *HFS*, *IFS*, *LFS*, *RES* and the raw production sequence is shown in Fig. 9.

Hilbert-Huang Transform (HHT) is applied on the raw sequence, *HFS*, *IFS*, *LFS* and *RES* to obtain their hilbert spectrum. It is obvious that both the raw and the reconstructed sequence exhibit significant production variations at the initial stage, with rapid changes occurring over a short period. The raw sequence corresponds to an intermediate-frequency, high-amplitude sequence, as shown in Fig. 10a. *HFS* represents a high-frequency, intermediate-amplitude sequence, as shown in Fig. 10b. *IFS* is categorized as an intermediate-frequency, low-amplitude sequence, as shown in Fig. 10c. *LFS* belongs to the low-frequency low-amplitude sequence, as shown in Fig. 10d. *RES* corresponds to a low-frequency, intermediate-amplitude sequence, as shown in Fig. 10e. Each reconstructed sequences exhibits distinct characteristics, forming the basis for subsequent forecasts. From the Hilbert spectrum, it is observed that *HFS* has a relatively high frequency, averaging 2.5×10^{-6} Hz (equivalent to 4.6 days), with an amplitude reaching 300 t/d. This fluctuation may result from frequent manual well-switching, which significantly impacts production (Fig. 10b). *IFS* has a frequency of approximately 0.8×10^{-6} Hz (equivalent to 14.4 days) and a maximum amplitude of 30 t/d, likely associated with fine-tuning of the oil well production system, such as bottom-hole pressure adjustments (Fig. 10c). *LFS* exhibits a much lower frequency of around 10^{-8} Hz (equivalent to 1000 days) and a maximum amplitude of 10 t/d, possibly influenced by the commissioning of nearby wells or production adjustments (Fig. 10d).

4.3. Production forecast

The LSTM forecast results of *HFS*, *IFS*, *LFS*, and DCA forecast result of *RES* are shown in Fig. 11. The forecast accuracy of LSTM varies significantly across different reconstruction sequences. The performance of LSTM is relatively poor for *HFS*, which has high frequency and amplitude (Fig. 11a). In contrast, *IFS* and *LFS*, which have lower frequencies, exhibit better forecast performance (Fig. 11b and c). The points in the true-forecasted value intersection diagram are closely distributed along the 45° line, and the distribution histograms of the true-forecasted value align well. Among them, *LFS* demonstrates the best forecast performance. The forecast error indicates that *HFS* has the largest forecast error on the test set, with an RMSE of 1.657. The forecast error on *IFS* test set is relatively lower (RMSE = 0.373), while LFS exhibits the smallest error (RMSE = 0.026). Other evaluation parameters and the performance of LSTM on the training set are shown in Table 2.

The *RES* value fitted by DCA are slightly lower than the true production, but the error remains within 1 t/d, which is within an acceptable range. The intersection diagram shows that values are closely aligned with the 45° line (Fig. 11d), where RMES is 11.777 and MEA is −0.002.

The final well production forecast is obtained by combining the four forecasted components, as illustrated in Fig. 11e. The proposed model effectively captures production fluctuations, with true-forecasted intersection points concentrat near the 45° line. The overall RMSE is 1.743, and the distribution histograms of the true value and forecasted value are in good agreement.

This research also analyzes the performance of four other machine learning models: a single LSTM, BP neural network, RF, and CNN. As shown in Fig. 12a, there are significant performance differences across models on the training set. CNN performs the best, achieving an RMSE of 0.172, far lower than other models. The second-best performer is RF, with an RMSE of 1.941. The proposed model demonstrates moderate

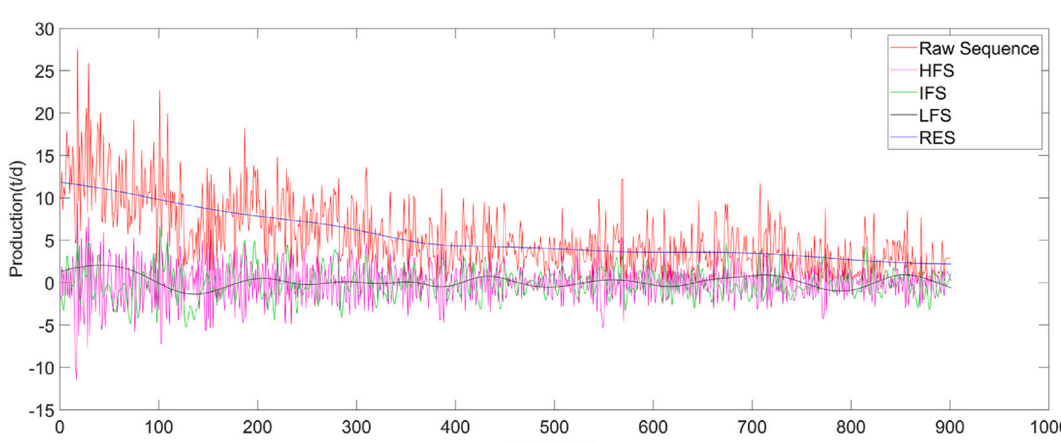


Fig. 9. Reconstruction of well production sequence in tight reservoir based on clustering results.

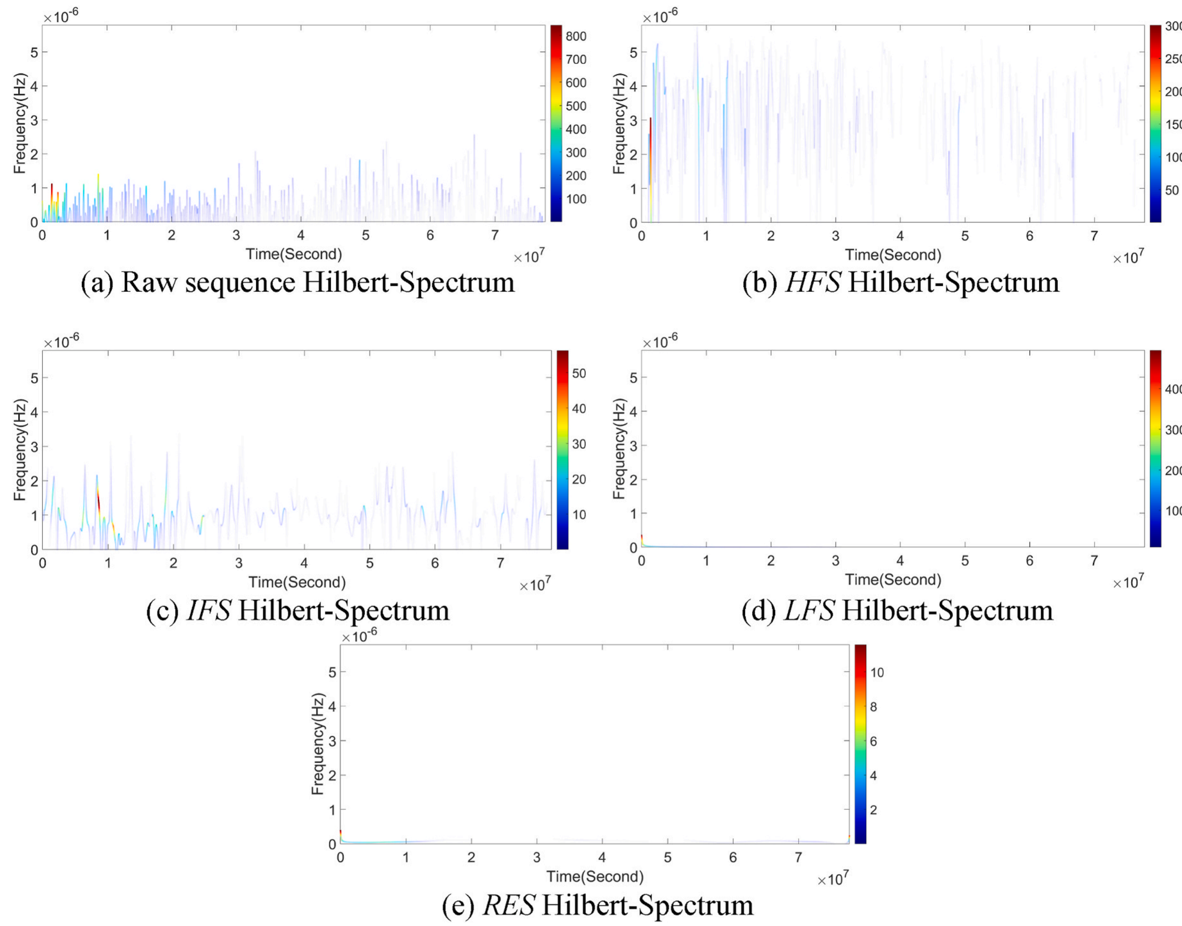


Fig. 10. Hilbert spectrum of different decomposition-reconstruction sequences.

performance, with an RMSE of 2.273, while LSTM and BP perform relatively poorly (Fig. 12c).

From the results of test set, the proposed model demonstrates the best forecast accuracy (Fig. 12b), with an RMSE of 1.743. Compared to a single LSTM model (RMSE = 2.328), the error is reduced by 25 %. From the perspective of MAE, MSE, and MAPE, the proposed model improves accuracy by at least 28 %, 37 %, and 69 %, respectively (Fig. 12d), indicating superior generalization ability. In contrast, CNN that performs best on the training set performs the worst on the test set, reflecting the poor generalization ability. The results in Table 3 provide further insights into model errors across training and test sets.

The approach taken in this research effectively mitigates the impact of sequence nonlinearity and non-stationarity. Compared with single LSTM, BP, RF, CNN model, the proposed model significantly improves the accuracy of production forecast of tight oil fractured horizontal wells.

5. Discussion

5.1. Findings and theoretical implications

Finding 1: EMD effectively separates the declining trend in the production sequence. Tight oil fractured horizontal well production consists of natural decline and fluctuation, which complicates machine learning-based forecast. This research successfully extracted these distinct subsequences using the EMD method, providing a solid foundation for subsequent forecasting.

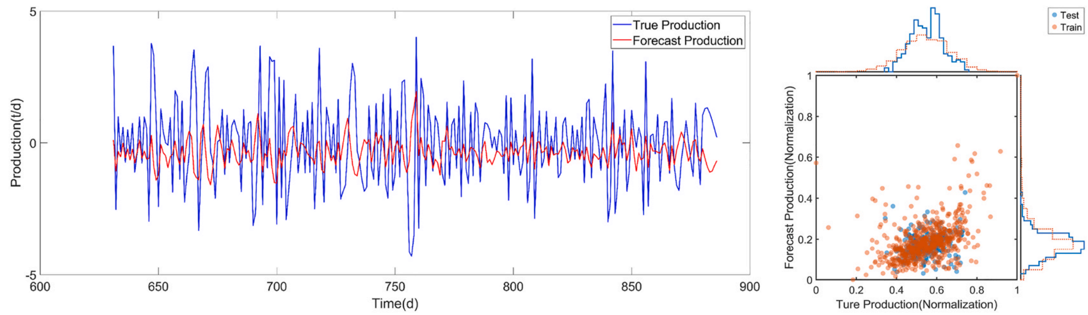
Finding 2: The data-driven model constrained by physical knowledge exhibits greater interpretability. Due to the absence of a mathematical model for tight reservoir, empirical DCA is used to fit the natural decline component of the production sequence, while LSTM forecasts the fluctuating component. Introducing physical knowledge constraints enhances the model's forecast performance. At the same time, it also provides a new interpretable idea for the prediction of oil well production and other sequences with long-term trend.

Finding 3: Cluster analysis of *IMF* ensures high forecast accuracy while reducing costs. EMD decomposition generates multiple *IMFs* and one *RES*. However, an excessive number of *IMFs* increases computational time and cost. This research confirms that *ApEn*-based clustering groups sequences with similar characteristics, optimizing resource efficiency while maintaining forecast accuracy.

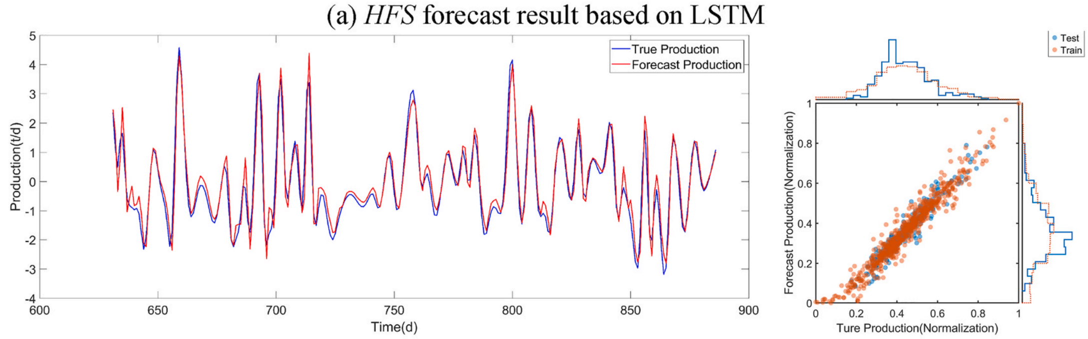
Finding 4: Signal processing techniques show promising applications in production sequence analysis. The *EMD*, *ApEn*, and *HHT* methods employed in this research originate from signal processing. Results indicate that these techniques effectively analyze production sequences, demonstrating their potential for characterizing production trends and improving forecasting accuracy.

5.2. Practical implications and suggestions

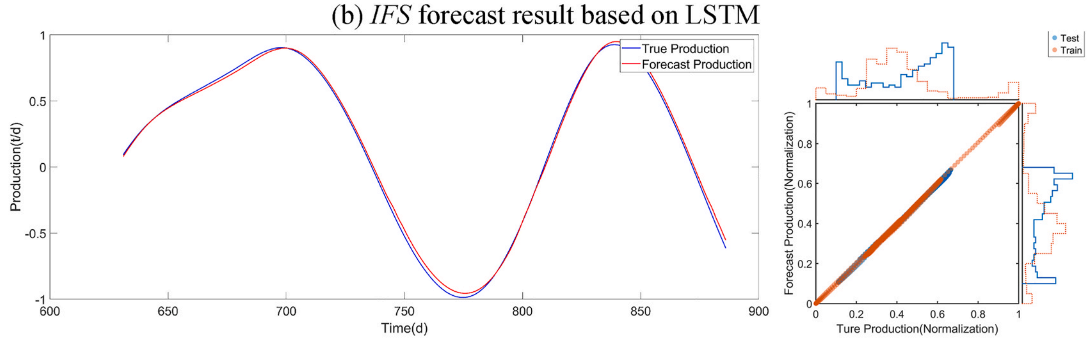
This research proposes a signal processing based model constrained by physical knowledge for forecasting the production of horizontal wells in tight reservoirs. For oil fields, accurate production forecast is the basis for evaluating reservoir production capacity, formulating oil field development plans, and adjusting production measures. For researchers, the signal processing and physical constraint methods introduced in this research provide new ideas for the analysis of other similar time series.



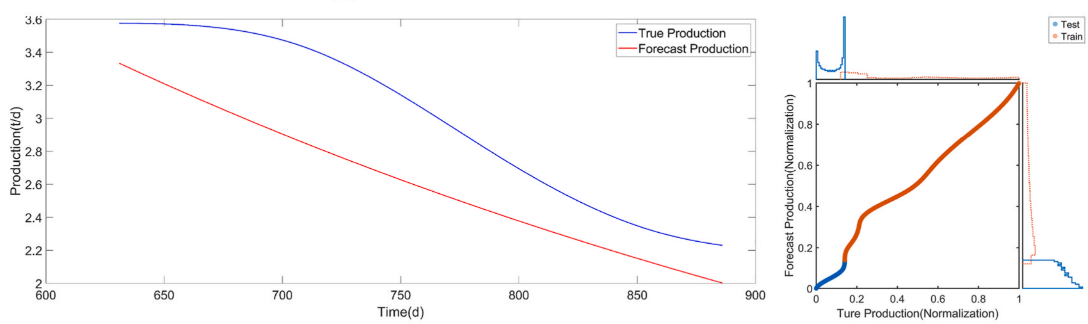
(a) *HFS* forecast result based on LSTM



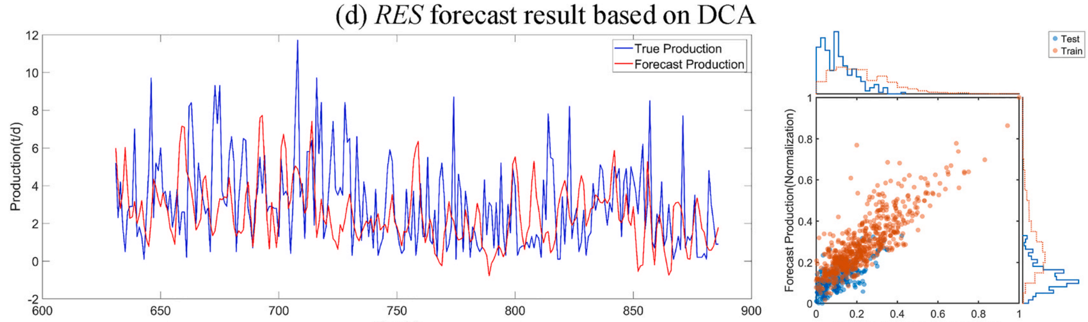
(b) *IFS* forecast result based on LSTM



(c) *LFS* forecast result based on LSTM



(d) *RES* forecast result based on DCA

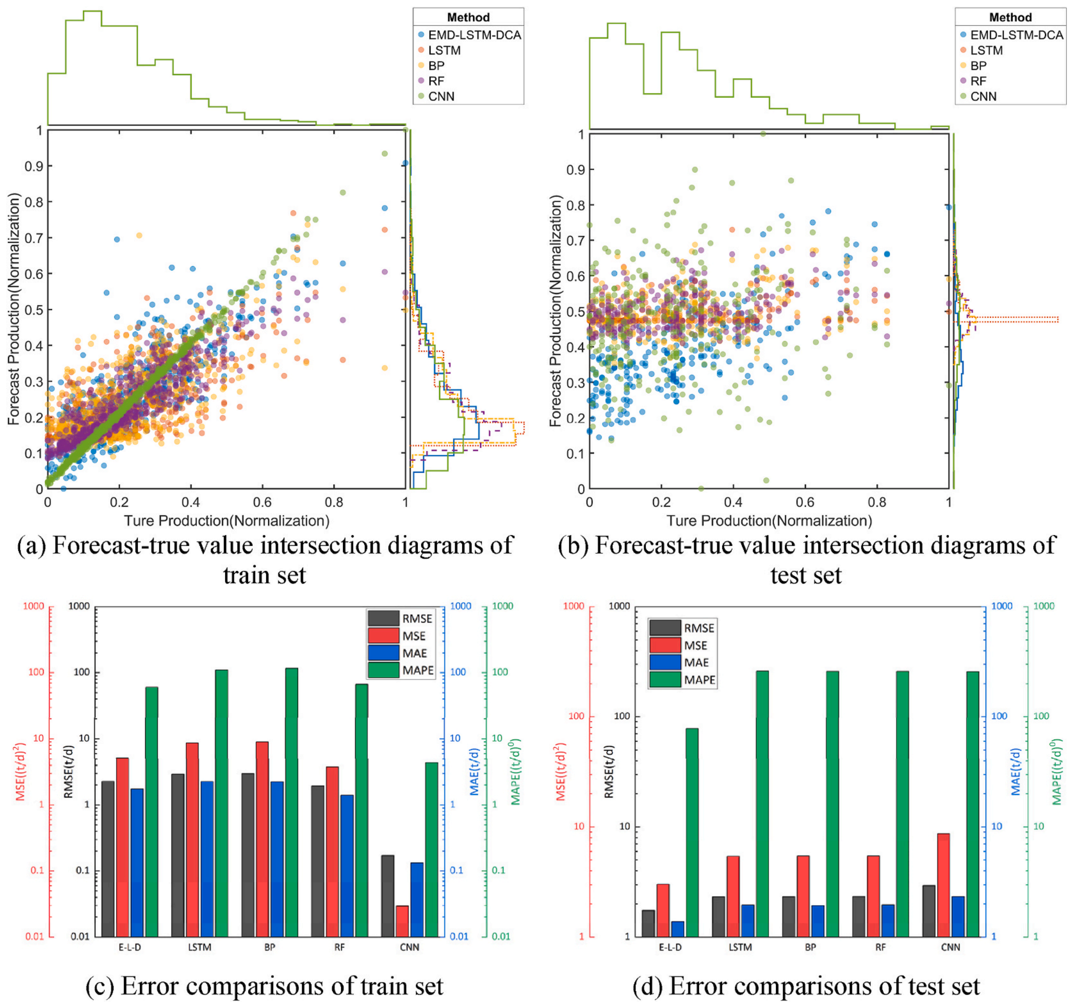


(e) Synthetic Sequence (SS) forecast result

Fig. 11. LSTM/DCA forecast results for each sequence.

Table 2
LSTM/DCA forecast error for each sequence.

Accuracy	RMSE		MSE		MAE		MAPE	
	Train	Test	Train	Test	Train	Test	Train	Test
HFS	2.284	1.657	5.218	2.744	1.740	1.340	170.442	250.501
IFS	0.469	0.373	0.220	0.139	0.347	0.285	160.485	227.921
LFS	0.024	0.026	0.001	0.001	0.019	0.022	28.937	23.970
RES	0.426	0.410	0.182	0.168	0.306	0.386	6.259	12.582
SS	2.273	1.743	5.165	3.038	1.748	1.379	60.716	78.285



(a) Forecast-true value intersection diagrams of train set
(b) Forecast-true value intersection diagrams of test set
(c) Error comparisons of train set
(d) Error comparisons of test set

Fig. 12. Forecast-true value intersection diagrams and error comparisons for different models.

Table 3
Forecast error comparison across different models.

Accuracy	RMSE		MSE		MAE		MAPE	
	Train	Test	Train	Test	Train	Test	Train	Test
E-L-D	2.273	1.743	5.165	3.038	1.748	1.379	60.716	78.285
LSTM	2.943	2.328	8.660	5.421	2.258	1.957	110.255	260.455
BP	2.997	2.335	8.980	5.451	2.238	1.929	116.583	258.411
RF	1.941	2.338	3.767	5.466	1.405	1.964	67.035	259.101
CNN	0.172	2.949	0.030	8.698	0.132	2.333	4.381	257.331

6. Conclusion

The nonlinear and non-stationary nature of production data, along with external interferences, makes the forecast of tight oil wells a challenge in petroleum engineering. This research proposes an LSTM-DCA method to forecast the production of horizontal wells in tight reservoirs, which refrains the limitations of the traditional single model.

This research introduces the commonly used EMD in signal processing. Results indicate that EMD decompose the production of tight oil fractured horizontal wells into several *IMF* and a *RES*. *IMF* captures the fluctuating part of the production sequence that deviates from the ideal decline law caused by various factors, while the residual component represents the production decline trend.

ApEn is employed to measure the complexity of each *IMF*. Based on *ApEn*, K-means is used to divide *IMF* into three categories: *HFS, IFS* and *LFS*. Each reconstructed sequence exhibits distinct frequency and amplitude characteristics, which improves the forecast accuracy while reducing computational cost. Compared with *HFS*, LSTM has a better forecast effect on *IFS* and *LFS*. Additionally, the *RES* extracted by EMD aligns well with the expected production decline trend.

Compared with single LSTM, BP, RF, CNN, the proposed hybrid model significantly enhances forecasting accuracy. The RMSE has been reduced by more than 25 %, while MAE has been reduced by more than 28 %. This improvement is attributed to LSTM's ability to capture small fluctuations and DCA's capability to model the production decline trend. The proposed low-cost, high-interpretability workflow can be conveniently applied in digital-oilfield platforms, enabling operators to make faster, more confident development decisions in tight-reservoir development.

6.1. Limitation

The forecast performance of the model proposed in this research emphasizes the importance of physical meaning and interpretability for production forecast, which can more effectively and accurately solve the engineering problems involved in this research. However, there are still some issues that have not been fully resolved:

The dataset used in this research is only depleted well production data during the development process, which is relatively clean compared to single well production data under complex operating conditions in tertiary oil recovery processes. However, compared with the research of simple models, it still proves the superiority of the proposed model.

The model proposed in this article includes three parts: mathematical model, empirical formula, and machine learning method. It is relatively complex to use. This research attempts to use a single model for production forecast, but the forecast effect is not ideal due to the decrease in production caused by natural energy depletion.

6.2. Future works

Future work may involve more complex physical knowledge as constraints on production. Complex conditions such as changes in displacement plugs and velocity should be considered the forecast of production, in order to explore more universal and interpretable forecast methods.

CRediT authorship contribution statement

Jinxin Cao: Writing – review & editing, Writing – original draft, Methodology, Data curation. **Yiqiang Li:** Supervision, Project administration, Funding acquisition. **Yaqian Zhang:** Writing – review & editing, Validation. **Xuechen Tang:** Visualization, Resources. **Qihang Li:** Writing – review & editing, Visualization, Investigation. **Yuling Zhang:** Investigation, Data curation. **Zheyu Liu:** Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This research is grateful for the National Natural Science Foundation of China (Grant 52074318), and for the pointers and suggestions given by the co-author and experts.

Data availability

Data will be made available on request.

References

Abdrabou, Moataz K., Han, Xue, Zeng, Yimin, Zheng, Ying, 2025. Recent developments in the utilization of unconventional resources: a focus on partial upgrading techniques and sustainability of Canadian Oil sand bitumen. Resour. Chem. Mater. 4, 100084.

Bhattacharya, Shuvajit, Ghahfarokhi, Payam Kavousi, Carr, Timothy R., Scott, Pantaleone, 2019. Application of predictive data analytics to model daily hydrocarbon production using petrophysical, geomechanical, fiber-optic, completions, and surface data: a case study from the Marcellus Shale, North America. J. Petrol. Sci. Eng. 176, 702–715.

Cao, Chong, Jia, Pin, Cheng, Linsong, Jin, Qingshuang, Qi, Songchao, 2022. A review on application of data-driven models in hydrocarbon production forecast. J. Petrol. Sci. Eng. 212, 110296.

Chahar, Jaiyesh, Verma, Jayant, Vyas, Divyanshu, Goyal, Mukul, 2022. Data-driven approach for hydrocarbon production forecasting using machine learning techniques. J. Petrol. Sci. Eng. 217, 110757.

Chalmond, B., Girard, S.C., 1999. Nonlinear modeling of scattered multivariate data and its application to shape change. IEEE Trans. Pattern Anal. Mach. Intell. 21, 422–432.

Cheng, C., Bai, E.W., 2021. Variable selection according to goodness of fit in nonparametric nonlinear system identification. IEEE Trans. Automat. Control 66, 3184–3196.

Das, Akash, Sudhanshu, Rahul, Shukla, Gaurav, Chaturvedi, Nitin Dutt, 2025. 'Artificial intelligence-based approach for water-cut prediction in crude oil processing'. Eng. Appl. Artif. Intell. 142, 109892.

Davtyan, Aram, Rodin, Alexander, Muchnik, Ilya, Romashkin, Alexey, 2020. Oil production forecast models based on sliding window regression. J. Petrol. Sci. Eng. 195, 107916.

Domala, Vamshikrishna, Kim, Tae-wan, 2023. Application of Empirical Mode Decomposition and Hodrick Prescot filter for the prediction single step and multistep significant wave height with LSTM. Ocean Engineering 285, 115229.

EIA, 2021. Annual energy outlook 2021 (AEO2021). July, 20. https://www.eia.gov/outlooks/archive/aeo21/.

Fan, Chengkai, Zhang, Na, Jiang, Bei, Liu, Wei Victor, 2024. Using deep neural networks coupled with principal component analysis for ore production forecasting at open-pit mines. J. Rock Mech. Geotech. Eng. 16, 727–740.

Fan, Dongyan, Sun, Hai, Yao, Jun, Zhang, Kai, Yan, Xia, Sun, Zhixue, 2021. Well production forecasting based on ARIMA-LSTM model considering manual operations. Energy (Calg.) 220, 119708.

Feng, J., Liu, Z., He, X., Li, Q., He, W., 2023. Vibration suppression of a high-rise building with adaptive iterative learning control. IEEE Transact. Neural Networks Learn. Syst. 34, 4261–4272.

Fragalla, Mandella Ali M., Yan, Wei, Deng, Jingen, Xue, Liang, Hegair, Fathelrahman, Zhang, Wei, Li, Guangcong, 2025. Time series production forecasting of natural gas based on transformer neural networks. Geoenergy Sci. Eng. 248, 213749.

Gonzalez-Zapata, Maritza, Astrid, de la Fraga, Luis Gerardo, Ovilla-Martinez, Brisbane, Tlelo-Cuautle, Esteban, Cruz-Vega, Israel, 2023. Enhanced FPGA implementation of Echo State Networks for chaotic time series prediction. Integration 92, 48–57.

Hakkoum, Hajar, Ali, Idri, Abnane, Ibtissam, 2024. Global and local interpretability techniques of supervised machine learning black box models for numerical medical data. Eng. Appl. Artif. Intell. 131, 107829.

Hornero, R., Aboy, M., Abasolo, D., McNames, J., Goldstein, B., 2005. Interpretation of approximate entropy: analysis of intracranial pressure approximate entropy during acute intracranial hypertension. IEEE (Inst. Electr. Electron. Eng.) Trans. Biomed. Eng. 52, 1671–1680.

Huang, N.E., Shen, Z., Long, S.R., Wu, M.C., Shih, H.H., Zheng, Q., Yen, N.C., Tung, C.C., Liu, H.H., 1998. The empirical mode decomposition and the Hilbert spectrum for nonlinear and non-stationary time series analysis. Proc. R. Soc. London, Ser. A 454.

Jongkittinarukorn, K., Last, N.C., Kabir, C.S., 2023. 'Decline-curve analysis with decline rate for improving reserves estimation in gas reservoirs'. Gas Sci. Eng. 116, 205044.

Kumar, Indrajeet, Tripathi, Bineet Kumar, Singh, Anugrah, 2023. Attention-based LSTM network-assisted time series forecasting models for petroleum production. Eng. Appl. Artif. Intell. 123, 106440.

Li, Xuechen, Ma, Xinfang, Xiao, Fengchao, Xiao, Cong, Wang, Fei, Zhang, Shicheng, 2022. A physics-constrained long-term production prediction method for multiple fractured wells using deep learning. J. Petrol. Sci. Eng. 217, 110844.

Ling, W., Yu, G., Li, Z., 2019. Lower limb exercise rehabilitation assessment based on artificial intelligence and medical big data. IEEE Access 7, 126787–126798.

Myshakin, Evgeniy M., Taiwo, Ajayi, Anderson, Brian J., Seol, Yongkoo, Ray, Boswell, 2016. Numerical simulations of depressurization-induced gas production from gas hydrates using 3-D heterogeneous models of L-Pad, Prudhoe Bay Unit, North Slope Alaska. J. Nat. Gas Sci. Eng. 35, 1336–1352.

Nadji-Tehrani, M., Eslami, A., 2020. A brain-inspired framework for evolutionary artificial general intelligence. IEEE Transact. Neural Networks Learn. Syst. 31, 5257–5271.

Najafzadeh, Mohammad, Mahmoudi-Rad, Mohammad, 2024. New empirical equations to assess energy efficiency of flow-dissipating vortex dropshaft. Eng. Appl. Artif. Intell. 131, 107759.
Ng, Cuthbert Shang Wui, Ghahfarokhi, Ashkan Jahanbani, Amar, Menad Nait, 2022. Well production forecast in Volve field: application of rigorous machine learning techniques and metaheuristic algorithm. J. Petrol. Sci. Eng. 208, 109468.
Ning, Yanrui, Kazemi, Hossein, Tahmasebi, Pejman, 2022. A comparative machine learning study for time series oil production forecasting: ARIMA, LSTM, and Prophet. Comput. Geosci. 164, 105126.
Pincus, S, M., 1991. Approximate entropy as a measure of system complexity. Proc. Natl. Acad. Sci. U. S. A 88, 2297–2301.
Sharma, P., Salman, M., Reza, Z., Kabir, C.S., 2019. Probing the roots of Arps hyperbolic relation and assessing variable-drive mechanisms for improved DCA. J. Petrol. Sci. Eng. 182, 106288.
Shi, Juntai, Huang, Liang, Li, Xiangfang, Sepehrnoori, Kamy, 2015. Production forecasting of gas condensate well considering fluid phase behavior in the reservoir and wellbore. J. Nat. Gas Sci. Eng. 24, 279–290.
Tugan, Murat Fatih, Weijermars, Ruud, 2020. Variation in b-sigmoids with flow regime transitions in support of a new 3-segment DCA method: improved production forecasting for tight oil and gas wells. J. Petrol. Sci. Eng. 192, 107243.
Wang, Muming, Hui, Gang, Pang, Yu, Wang, Shuhua, Chen, Shengnan, 2023. 'Optimization of machine learning approaches for shale gas production forecast'. Geoenergy Sci. Eng. 226, 211719.
Wang, Shuhua, Chen, Shengnan, 2019. Insights to fracture stimulation design in unconventional reservoirs based on machine learning modeling. J. Petrol. Sci. Eng. 174, 682–695.
Wang, Y., Kinsner, W., Zhang, D., 2009. Contemporary cybernetics and its facets of cognitive informatics and computational intelligence. IEEE Trans. Syst. Man Cybern. B Cybern. 39, 823–833.
Wang, Zhikai, Cheng, Linsong, Hamdi, Hamidreza, Jia, Pin, Cao, Renyi, Clarkson, Christopher R., 2023. A semi-analytical model for quantifying the inter-well communication in water-bearing shale gas-condensate reservoirs. Geoenergy Sci. Eng. 228, 211997.
Wang, Ting, Liu, Gang, 2009. An improved method to solve the end effect of EMD and its application on vibration signal. In: 2009 International Conference on Mechatronics and Automation, pp. 3977–3981.
Wen, Shuqin, Wei, Bing, You, Junyu, He, Yujiao, Xin, Jun, Varfolomeev, Mikhail A., 2023. Forecasting oil production in unconventional reservoirs using long short term memory network coupled support vector regression method: a case study. Petroleum 9, 647–657.
Wu, Yonghui, Cheng, Linsong, Huang, Shijun, Fang, Sidong, Jia, Pin, Xiang, Rao, 2019. An analytical model for analyzing the impact of fracturing fluid-induced formation damage on rate transient behavior in tight formations. J. Petrol. Sci. Eng. 179, 513–525.
Yan, Li, Wen, Hu, Wang, Zhenping, Jin, Yongfei, Guo, Jun, Liu, Yin, Fan, Shixing, 2025. Prediction and evaluation of key parameters in coalbed methane pre-extraction based on transformer and inversion model. Eng. Appl. Artif. Intell. 139, 109661.
Yan, Shuli, Peng, Mengna, Wu, Lifeng, Xiong, Pingping, 2025. A novel structural adaptive seasonal grey Bernoulli model in natural gas production forecasting. Eng. Appl. Artif. Intell. 148, 110407.
You, Lijun, Xie, Yunsong, Chen, Yang, Kang, Yili, Huang, Chen, Zhai, Huaijian, 2025. Development directions of formation damage evaluation and fracturing fluids in tight and shale oil reservoirs. Petroleum 11, 125–142.
Zha, Wenshu, Liu, Yuping, Wan, Yujin, Luo, Ruilan, Li, Daolun, Yang, Shan, Xu, Yanmei, 2022. Forecasting monthly gas field production based on the CNN-LSTM model. Energy (Calg.) 260, 124889.
Zhang, Bowei, Guo, Simao, Jin, Hui, 2022. Production forecast analysis of BP neural network based on Yimin lignite supercritical water gasification experiment results. Energy (Calg.) 246, 123306.
Zheng, W., Hu, J., 2023. Multivariate time series prediction based on temporal change information learning method. IEEE Transact. Neural Networks Learn. Syst. 34, 7034–7048.
Zhou, Guangzhao, Guo, Zanquan, Sun, Simin, Jin, Qingsheng, 2023. A CNN-BiGRU-AM neural network for AI applications in shale oil production prediction. Appl. Energy 344, 121249.